A

PROJECT REPORT

ON

"Mess management system"

A DISSERTATION SUBMITTED TO

M.S.B.T.E MUMBAI

For The Award of

DIPLOMA IN

COMPUTER ENGINEERING

BY

Mr. TEJAS A. CHUMBALKAR (ENROLLMENT:1610740076)

Mr. RUSHIKESH R. VASEKAR (ENROLLMENT:1610740067)

Mr. RUTURAJ B. KADAM (ENROLLMENT:1610740080)

Mr. PRASAD K. SAGAR (ENROLLMENT:1610740069)

UNDER THE GUIDANCE OF

Prof. PRASHANT S. BHANDARE

DEPARTMENT OF COMPUTER ENGINEERING

SVERI'S COLLEGE OF ENGINEERING(POLYTECHNIC), PANDHARPUR

413304

2018-2019

AFFILIATED TO

M.S.B.T.E. MUMBAI.

SVERI'S COLLEGE OF ENGINEERING(POLYTECHNIC), PANDHARPUR-413 304

DEPARTMENT OF COMPUTER ENGINEERING

GOPALPUR, PANDHARPUR-413 304

CERTIFICATE

This is certify that the project entitled

"SHREE DENTAL CARE WEBSITE"

submitted by

Mr. TEJAS A. CHUMBALKAR (ENROLLMENT:1610740076)

Mr. RUSHIKESH R. VASEKAR (ENROLLMENT:1610740067)

Mr. RUTURAJ B. KADAM (ENROLLMENT:1610740446)

Mr. PRASAD K. SAGAR (ENROLLMENT:1610740069)

is a record of bonafide work carried out by him, in the partial fulfilment of the requirement for the award of Diploma of Engineering (Computer Engineering) at SVERI'S COLLEGE OF ENGINEERING(POLYTECHNINC), PANDHARPUR under the University of MSBTE. This work is done during year 2018-2019, under our guidance.

Prof. Prashant S. Bhandare
(Project Guide)

Prof. Prashant S. Bhandare
(Project Coordinator)

Prof. Ameya S. Bhatlavande
(HOD, Dept. of CO)

Dr. Nitin D. Misal
(Principal)

External Examiner Name and Sign:

Place: Pandharpur

Date: / /

Acknowledgement

While working on the current dissertation of "Shree Dental Care" got exposed to various new technologies due to support of my guides teachers and friends. I would like to express my heartfelt gratitude to my Guide Prof. Prashant S. Bhandare who acted as a source of inspiration in all spheres of my dissertation phase and necessarily providing all resources along with a great platform to accomplish my target. I would like to give special thanks to our Head of Dept. Prof. Ameya S. Bhatlavande and Project coordinator Prof. Prashant S. Bhandare for giving valuable guidelines for completing this course.

I would like to acknowledge our Principal Dr.Nitin D. Misal sir for in-troducing the Diploma (Computer Engineering) course in our Institute. Otherwise, it would have been difficult for me to take admission for this course at any other place. I would also like to thank other faculty and staff members and non-teaching staff of my department. I would also like to thank Prof. Prashant S. Bhandare for teaching me to prepare Latex report. I would like to thank all my Diploma batch mates for their help in group learning process. I would like to thank my family and friends who helped me with technical and emotional guidance during this tenure. It will be worthful to close the acknowledgement section conveying a heartfelt thanks to all.

Tejas Appasaheb Chumbalkar

Rushikesh Ramesh Vasekar

Ruturaj Bhupesh Kadam

Prasad Kishor Sagar

DECLARATION

I hereby declare that the project entitled "Shree Dental Care" is completed and written by us for the award of DIPLOMA IN COMPUTER ENGINEERING, work under the guidance of Prof. Bhandare P.S at the SVERI's COE(Poly), Pandharpur. I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

I have taken care in all respect to honor the intellectual property right and have acknowledged the contribution of others for using them in academic purpose and further declare that in case of any violation of intellectual property right or copyright I as a candidate,will be fully responsible for the same. Our supervisor should not be held responsible for full or partial violation of copyright or intellectual property right.

Tejas Appasaheb Chumbalkar
Rushikesh Ramesh Vasekar
Ruturaj Bhupesh Kadam
Prasad Kishor Sagar

Place: Pandharpur
Date :

Abstract

This website is aimed at developing a Shree Dental Care Website that is of importance to dental clinic or relevant clinics. This system may be used for monitoring patient personal information and dental health information. This application is being developed for dental clinics to maintain and facilitate easy access of information to dentist or doctor. For this the dentist must login and register the patient information in the system after which they can access as well as modify data as per the requirements needed to them. In any Dental Clinics, it is very hard to remember all Information about all Patients such as Contact Number, medical history etc. Also Difficult to remember Patients' Data such as treatment done, payment details, next appointment etc. The Dental Clinic System web application involves Patients Contact list sorted with name, also includes medical history, treatment done and balance details which are stored in database, appointment details shows the details such as time and date appointed to the patient. As this is a web application the doctor can easily access the necessary details of each and every patient anywhere hence this becomes advantage of this web application

Contents

1 Introduction **1**

1.1 General Introduction 1

1.2 Background 2

2 Literature Review **3**

2.1 Existing System 4

2.1.1 Dental Information Management System(DIMS) 4

2.1.2 Shah Dental Clinic Web Application 4

2.1.3 Practo's Android Application 4

2.2 Proposed System 5

2.2.1 Shree Dental Care Website 5

3 Problem Statement **6**

3.1 Objective 6

3.2 Scope 7

3.3 Design Overview 8

3.3.1 Data flow diagrams 8

3.3.2 UML Diagram 12

4 Methodology **16**

4.1 System Model 16

4.2 Project Development 16

4.2.1 Phase 1: - Knowledge Acquisition 16

4.2.2 Phase 2: - Design 16

4.2.3 Phase 3: - Implementation 16

4.2.4 Phase 4: - Testing 16

4.2.5 Phase 5: - Documentation 17

4.3 Strategic Approach To Software Testing 17
4.3.1 Unit Testing 18
4.3.2 White box testing 18
4.3.3 Basic path testing 18
4.3.4 Conditional testing 18
4.3.5 Data flow testing 19
4.3.6 Loop testing 19

5 Implementation **20**

6 Conclusion **27**
6.1 Appendix 28

7 Experiment and Result Analysis **29**

List of Figures

3.1 Level 0 Data flow diagram 8
3.2 Level 1 Data flow diagram 9
3.3 Level 1.1 Doctor Module 10
3.4 Level 2 Dataflow Diagram 11
3.5 Use case Diagram 12
3.6 Sequence Diagram 13
3.7 Class Diagram 14
3.8 Activity Diagram for User 15

4.1 Testing Process 17

5.1 Home page of Our Web Application 20
5.2 Online Appointment Application Form. 21
5.3 About Us Page of Web Application. 22
5.4 Contact Us Page of Web Application. 23
5.5 Facilities Page of Web Application. 24
5.6 Visiting Doctors Page of Web Application. 25
5.7 Gallery of Page of Web Application 26

List of Tables

6.1 Appendix 28

Chapter 1

Introduction

1.1 General Introduction

The title of the project is "Shree Dental Care" is a web application that aims at providing a health medication and treatment for all types of dental patients.
Surely, everyday there are people that need to use the dental clinic services.But how can clinic provide a faster and efficient services if they are still using the traditional method on their daily operation? The traditional method means the customers need to fill in their detail in registration form manually and the information will only keep in files. After the registration, the files will be place in the rack and this will cause problems like taking a longer time to retrieve the information, make mistakes during writing or misplaced the files. Nowadays, technology has changed many aspects of life and people's daily life is becoming indivisible from the network due to the development of Internet.With dental clinic system, the process gets much faster and more efficient than traditional way.

Even though the dental clinic had a system, but they still do not use the application. It is because there are facing several problems during process management. So, people from outside cannot view their treatment examination that they have done in the dental clinic. There are also problem when the patient forgot their appointment time because they cannot check their appointment schedule through online. Although the appointment schedule was provided in the manual, it is possible that the paper will be missing.[1]
Dental Clinic System with Notification is specially designed to let the clinic staff has a high efficiency management tools, computerized and systematic patient's record, and detail of treatment records. This system also provides appointment feature, which allow staffs to view the appointment that already made by doctors and process it by sending a notification to patients. Patients will

receive the notification about their appointment details on their mobile phone. This new system will replace the current system that is used in clinic and surely this system will improve the clinic services and make their daily operation running smoothly.[2]

1.2 Background

Basically, Dental Clinic System developed to implement process in the dental clinic starting with patient registration until dentist or nurse settled the patient treatment. This system functions to implement process of the dental clinic that record about dentist profile, patient personal detail, patient treatment, patient appointment, patient billing, and dental treatment and disease. Patient personal detail consists the information about general information, previous medical history and previous dental history.[3][4] Patient treatment module consists the information about the tooth examination and record and list of treatment that has done.

Besides that, dental treatment and disease module provide information about the cause of tooth extraction and tooth filling. It is allowing the dentist or nurse to trace whether the patient needs to make tooth extraction or tooth filling. Decision Support System for Dental Clinic is an independence management solution for managing information in the non-government clinic. The propose system is also focuses to improve the current system in part of record patient tooth treatment.

We got one idea of the Desktop Application. That's why we developed the "Dental Clinic System Web Application". This is Web based application which helps to do get information that providing a health medication and treatment for all types of dental patients.

Chapter 2

Literature Review

Literature review refers on interview, book, internet to grab some information about this project to give clear outlook for developing this project. Current system of the dental clinic also has the system that can handle the process management as a computerized. But, there are some problems while using this system and that cause them use the manual system. So, this literature review has covered the process of manual system that cause the current system is difficult to use. Every organization needs a system that would manage their data and process. Usually they used the predictable method to manage and store their data.

In every organization keeping record is the vital task. For the clinic, it is very important to keep the patient record for any reference. There are two ways of keeping the record such as using the manual method or computerized method. The computerized system would be better than manual system for maintaining the record of the patients. Hence, by using this system there are number of advantages than the manual system.[3] One of the main advantages of using the computerized system is that it is not only easy, but it also saves the time to search the patient record. If they use manual system for fetching the patient record they have a hard time to find it. The computerized will give the opportunity for the dental clinic to do work more effective and efficiently if the clinic use it.[4]

2.1 Existing System

2.1.1 Dental Information Management System(DIMS)

DIMS is a windows based medical billing software designed for medical offices, chiropractors, dentists and medical billing services. Whether you are interested in starting your own home based medical billing service or manage a private practice, DIMS gives you the best software on the market. DIMS is all about saving you time, improving your cash flow, and helping you better manage the business side of your practice. With all of its improvements and new features, DIMS is having a disadvantage of messaging. So, it has been overcome in this Dental Clinic System Web Application

2.1.2 Shah Dental Clinic Web Application

SDC is a web Application which is used by us as reference for creating our Web Application. In this Web Application there is no feature of Message sending which is implemented in our system. Also from this application we take the references for the Patient Registration form, Estimation Details, Next Appointment, etc. In this system there is no feature which print the Prescription Report of the patient, this module is also implemented by us in our web Application.

2.1.3 Practo's Android Application

Practo Application is used by us for taking reference for the Making Online appointment of the patient. But in this Application there are no features of the registration page and many more pages required for doctor, this pages are implemented in our Web Application.

2.2 Proposed System

2.2.1 Shree Dental Care Website

Our website Dental Clinic System(DCS) is user friendly and dynamic.In this any user can search our website from anywhere and utilize it. He can utilize it in many ways such as if any end user is having a problem about his tooth then he/she can contact our doctor and send their problem by using the contact us page. Before visiting the clinic he/she can view are clinic as we have provided a clinic tour page in which we have displayed all the interior and exterior views of our clinic, they can also view our services, machines etc. The end user can also see about the visiting doctors of our clinic.

We also have provided that at the time of weekend what are the timings of the clinic and the way to book an special appointment. The special appointment will be taken only on weekends. By filling online form on our website the appointment can be booked. The appointment registration form includes the few personal details like 'Name', 'Contact No', 'Email-ID', 'Information about the problem of patient'. If there is an any major problem or issue regarding any disease/s, then the doctor or patient can communicate directly through the 'Email-ID' or 'Contact No'.

Chapter 3

Problem Statement

We are going to develop a websuite for"Shree Dental Care"

3.1 Objective

The Dental Clinic System web application contains all information about the Doctors and patient of dental clinic. In Dental Clinic System Web application is also include patient report.

1. No need to remember all information related to patient.
2. All patient information will available.
3. All doctor contact information will available.
4. All patient information and treatment record will available.
5. It is helpful for patients to contact doctor.

3.2 Scope

1. The doctor can view the details of patients.
2. The patient can get the information about the visiting doctors on site.
3. The patient can get the contact with doctor through contact us page.
4. To get the more information about the doctor, or visiting doctors the E-mail ID, Facebook and Twitter links are provided.
5. The website is user friendly easy to use.

3.3 Design Overview

3.3.1 Data flow diagrams

1. **LEVEL 0**

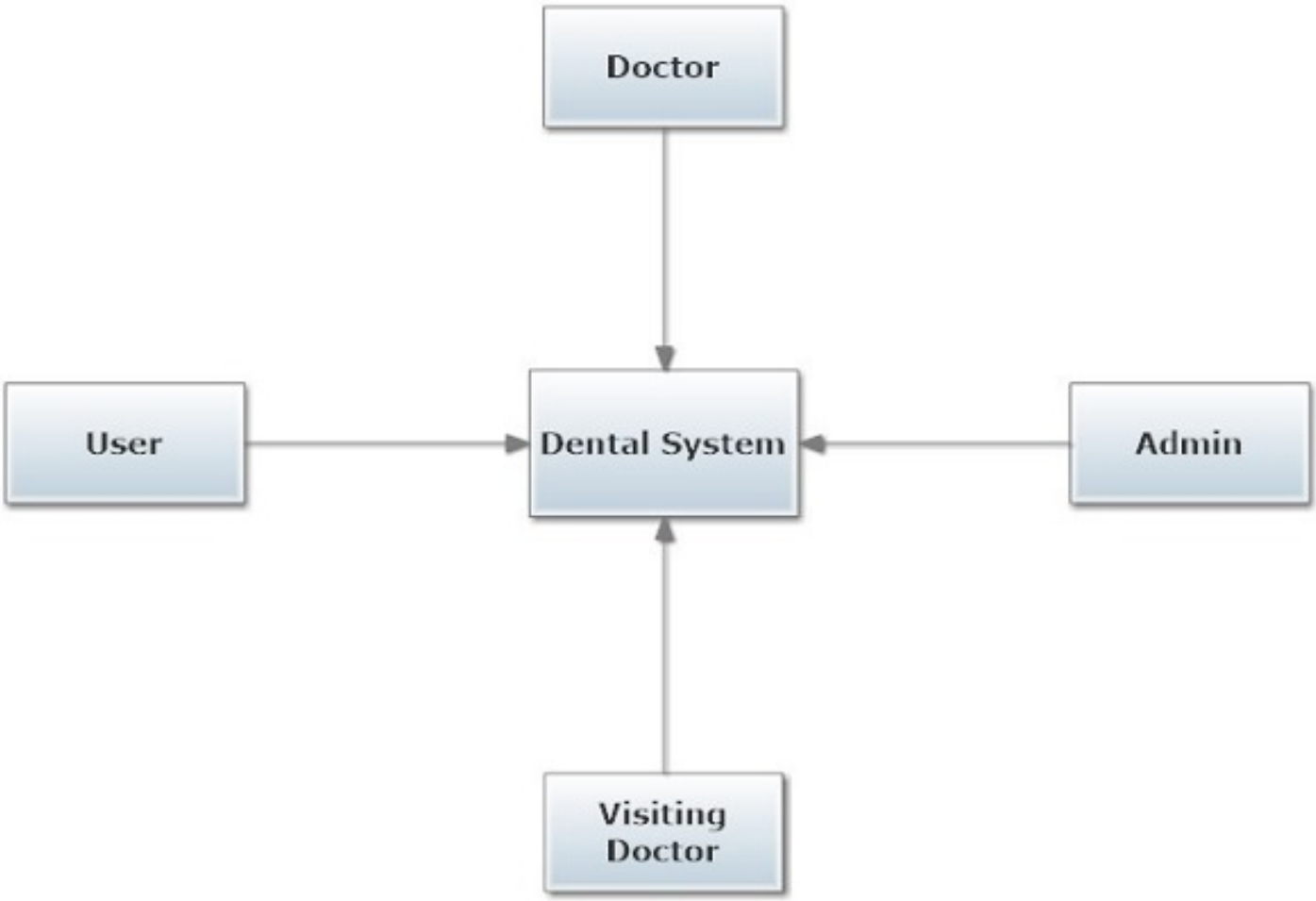

Figure 3.1: Level 0 Data flow diagram

2. **LEVEL 1**

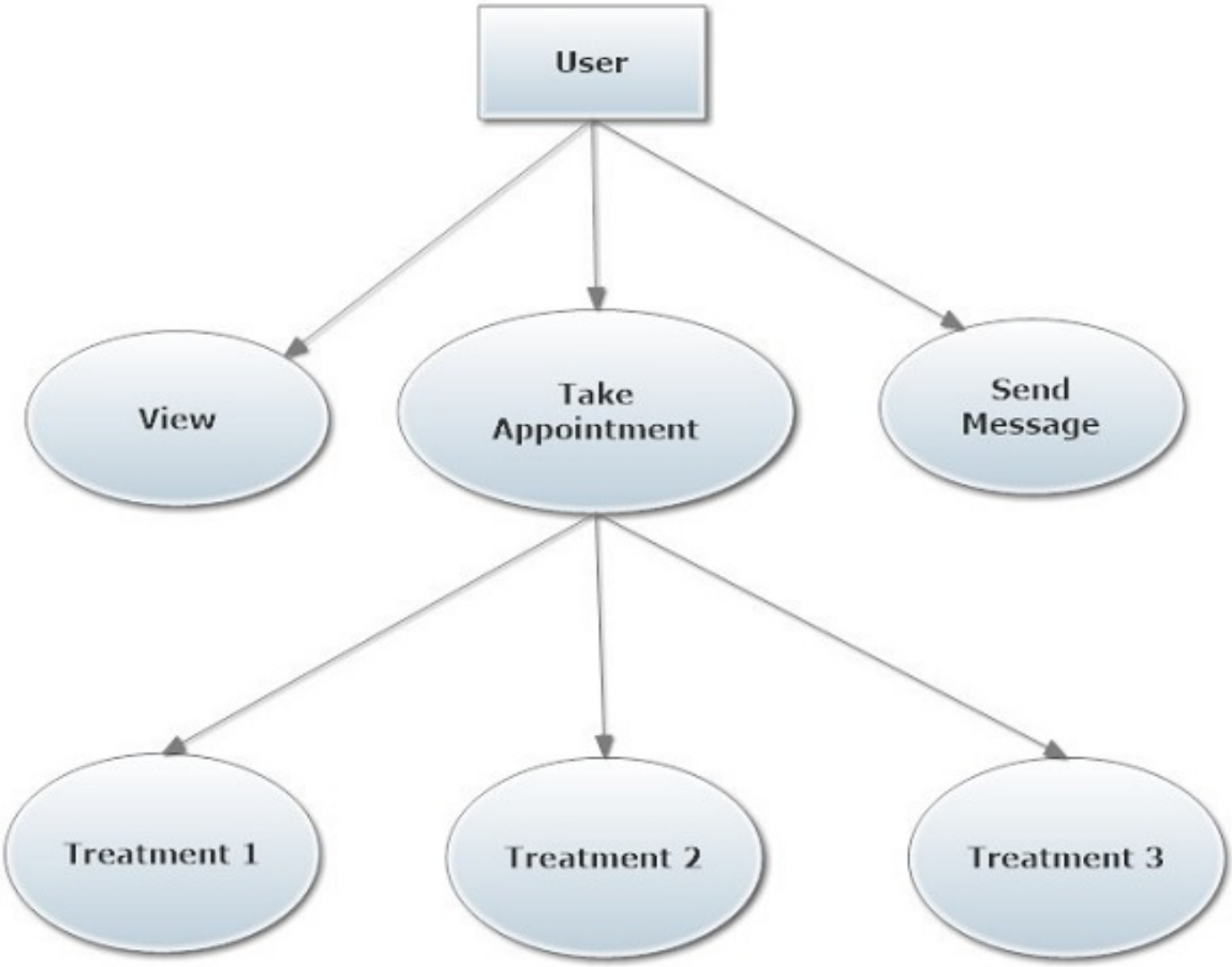

Figure 3.2: Level 1 Data flow diagram

3. **LEVEL 1.1**

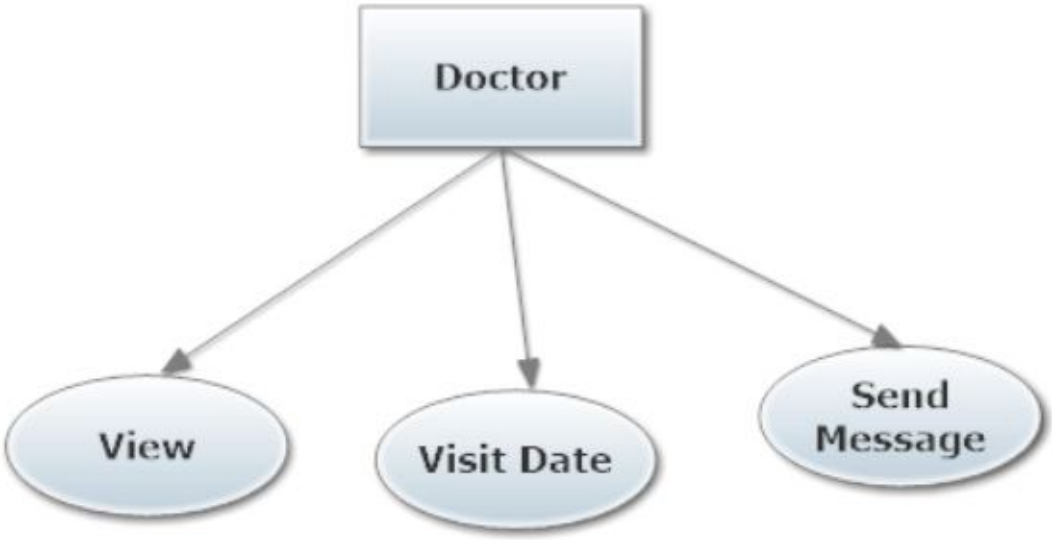

Figure 3.3: Level 1.1 Doctor Module

4. **LEVEL 2**

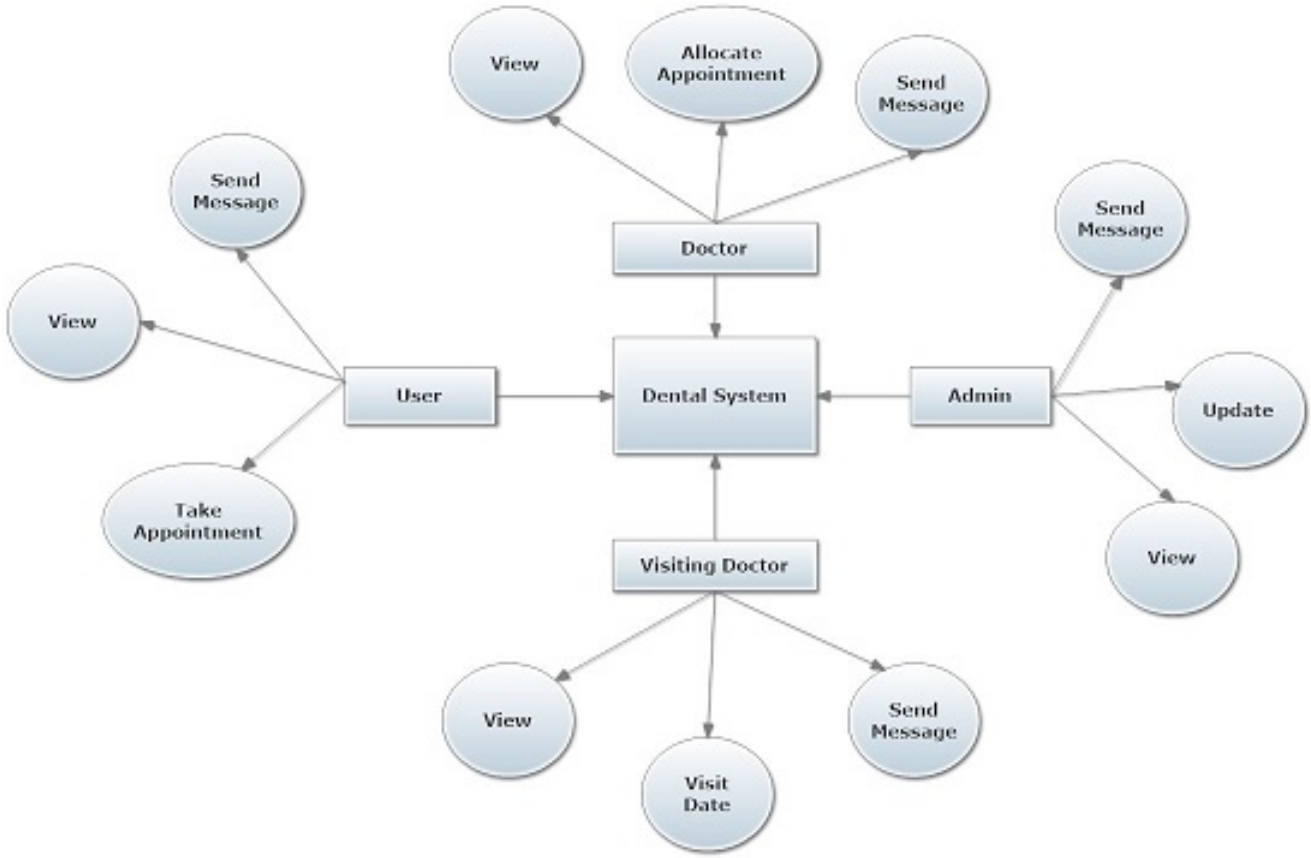

Figure 3.4: Level 2 Dataflow Diagram

3.3.2 UML Diagram

1. **Use case Diagram**

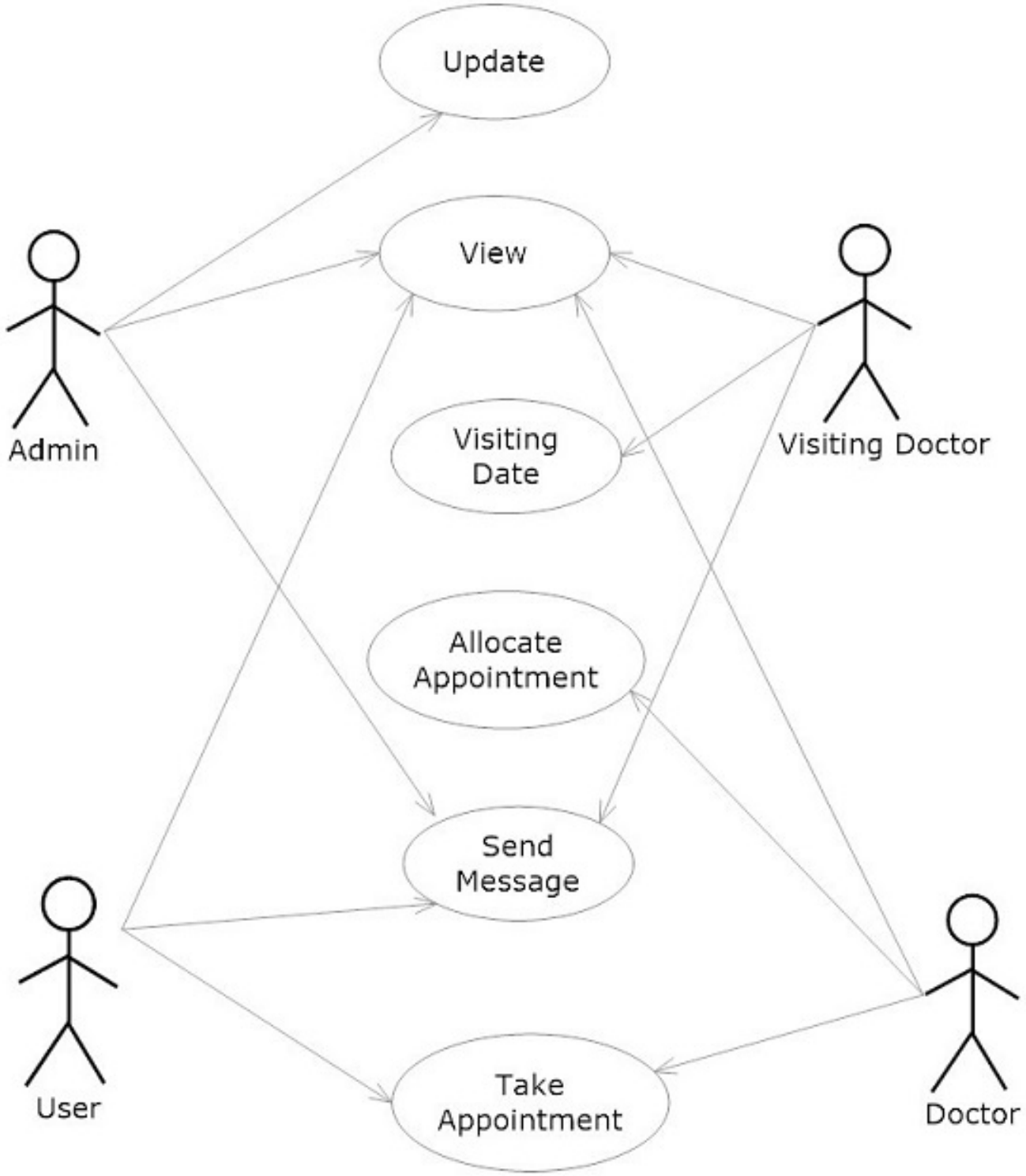

Figure 3.5: Use case Diagram

2. **Sequence Diagram**

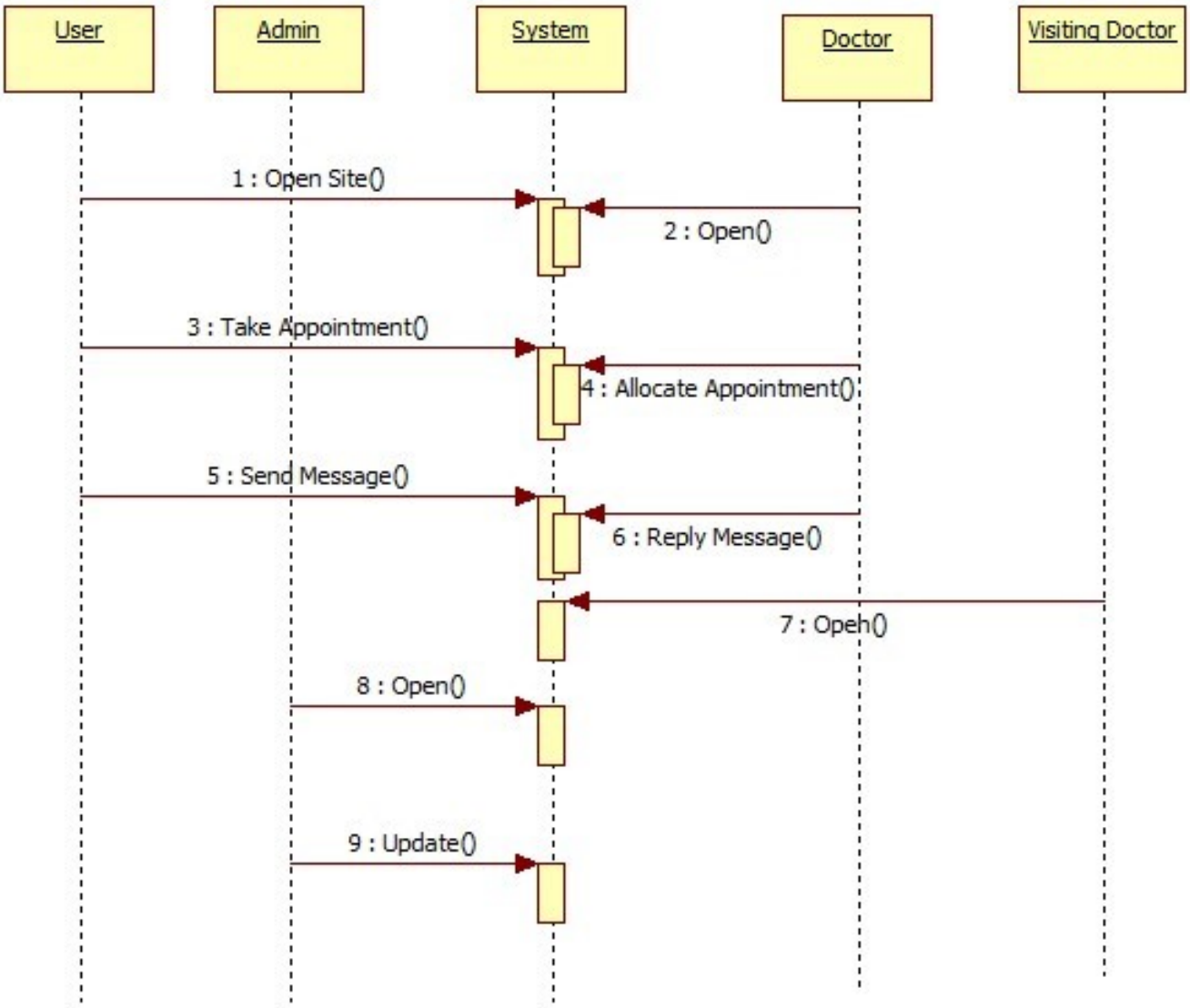

Figure 3.6: Sequence Diagram

3. **Class Diagram**

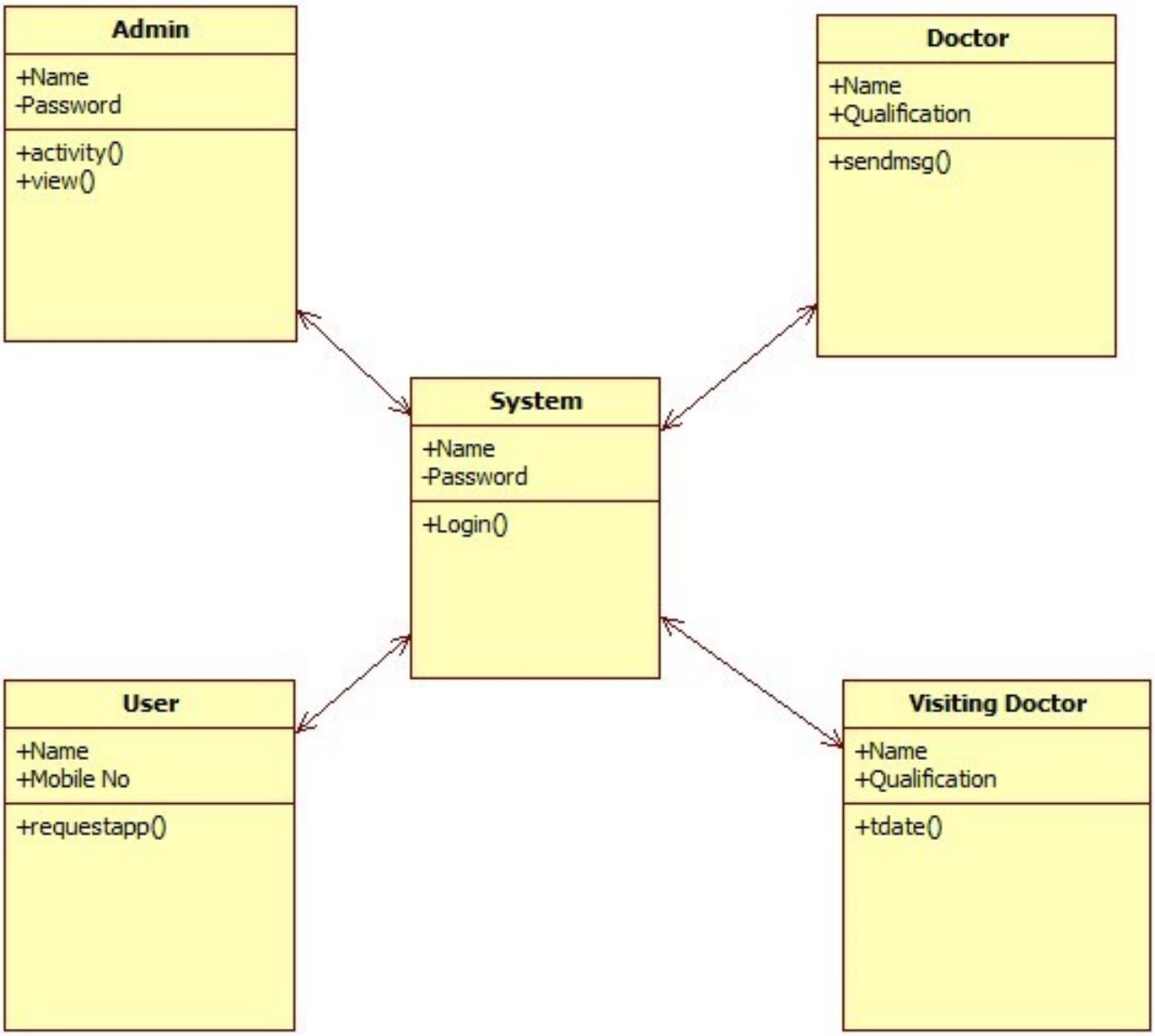

Figure 3.7: Class Diagram

4. **Activity Diagram**

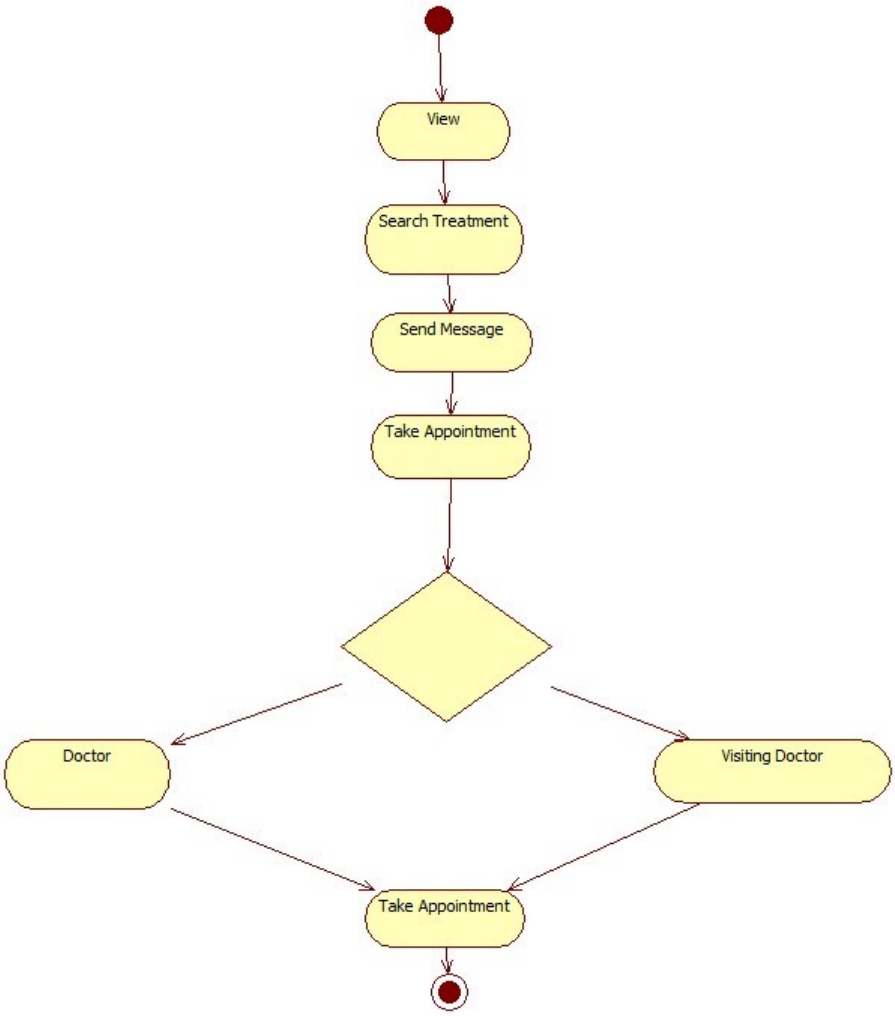

Figure 3.8: Activity Diagram for User

Chapter 4

Methodology

4.1 System Model

During the development of this project the team need to be understand the concept and the development of web applications embedded circuits.

4.2 Project Development

In project development Process there are five phases in which all project development is explained. Those five phases are as follows: -

4.2.1 Phase 1: - Knowledge Acquisition

In this phase we study about how to design programming application and for this purpose we collect data, information about it by observation and internet reading.

4.2.2 Phase 2: - Design

Second phase is design in which we will design the web architecture.

4.2.3 Phase 3: - Implementation

In this phase we will implement coding, by using Sublime Text Editor.

4.2.4 Phase 4: - Testing

In this phase we will go to test coding.

4.2.5 Phase 5: - Documentation

In this phase we will going to prepare final report.

4.3 Strategic Approach To Software Testing

The software engineering process can be viewed as a spiral. Initially system engineering defines the role of software and leads to software requirement analysis where the information domain, functions, behavior, performance, constraints and validation criteria for software are established.

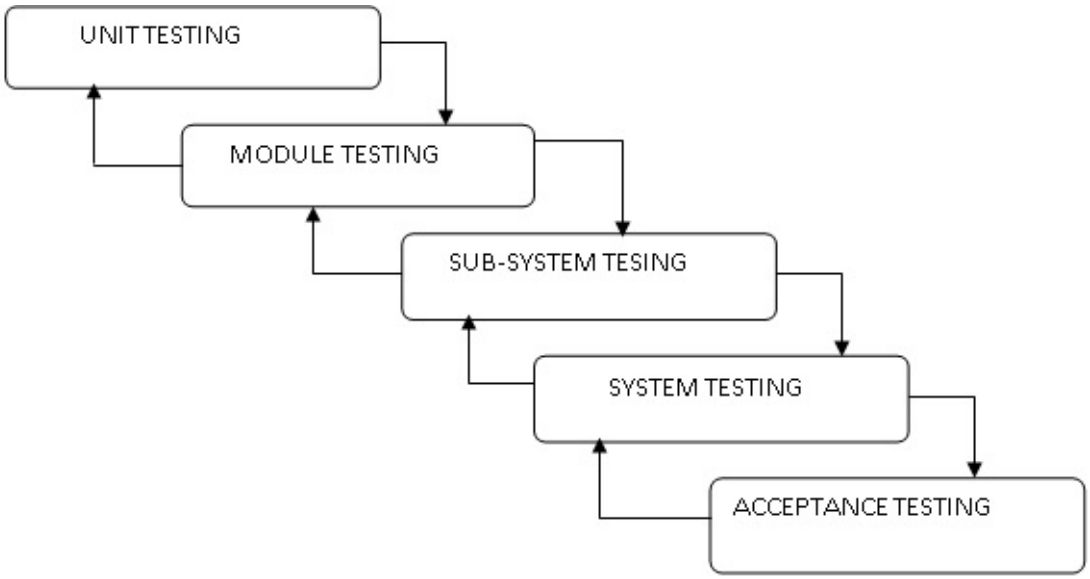

Figure 4.1: Testing Process

4.3.1 Unit Testing

Unit testing focuses verification effort on the smallest unit of software design, the module. The unit testing, we have is white box oriented and some modules the steps are conducted in parallel.

4.3.2 White box testing

This type of testing ensures that:

1. All independent paths have been exercised at least once
2. All logical decisions have been exercised on their true and false sides
3. All loops are executed at their boundaries and within their operational bounds
4. All internal data structures have been exercised to assure their validity.

4.3.3 Basic path testing

Established technique of flow graph with cyclomatic complexity was used to derive test cases for all the functions. The main steps in deriving test cases were: Use the design of the code and draw correspondent flow graph. Determine the cyclomatic complexity of resultant flow graph, using formula: V (G) =E-N+2 or V (G) =P+1 or V (G) =Number of Regions Where V (G) is cyclomatic complexity, E is the number of edges, N is the number of flow graph nodes, P is the number of predicate nodes. Determine the basis of set of linearly independent paths.

4.3.4 Conditional testing

In this part of the testing each of the conditions were tested to both true and false aspects. And all the resulting paths were tested. So that each path that may be generate on particular condition is traced to uncover any possible errors.

4.3.5 Data flow testing

This type of testing selects the path of the program according to the location of definition and use of variables. This kind of testing was used only when some local variable was declared. The definition-use chain method was used in this type of testing. These were particularly useful in nested statements.

4.3.6 Loop testing

In this type of testing all the loops are tested to all the limits possible. The following exercise was adopted for all loops:

(a) All the loops were tested at their limits, just above them and just below them.

(b) All the loops were skipped at least once.

(c) For nested loops test the inner most loop first and then work outwards.

(d) For concatenated loops the values of dependent loops were set with the help of connected loop.

(e) Unstructured loops were resolved into nested loops or concatenated loops and tested as above.

Chapter 5

Implementation

This chapter includes the various stages of project in the form of Graphical user interface.

i. Main Home Page

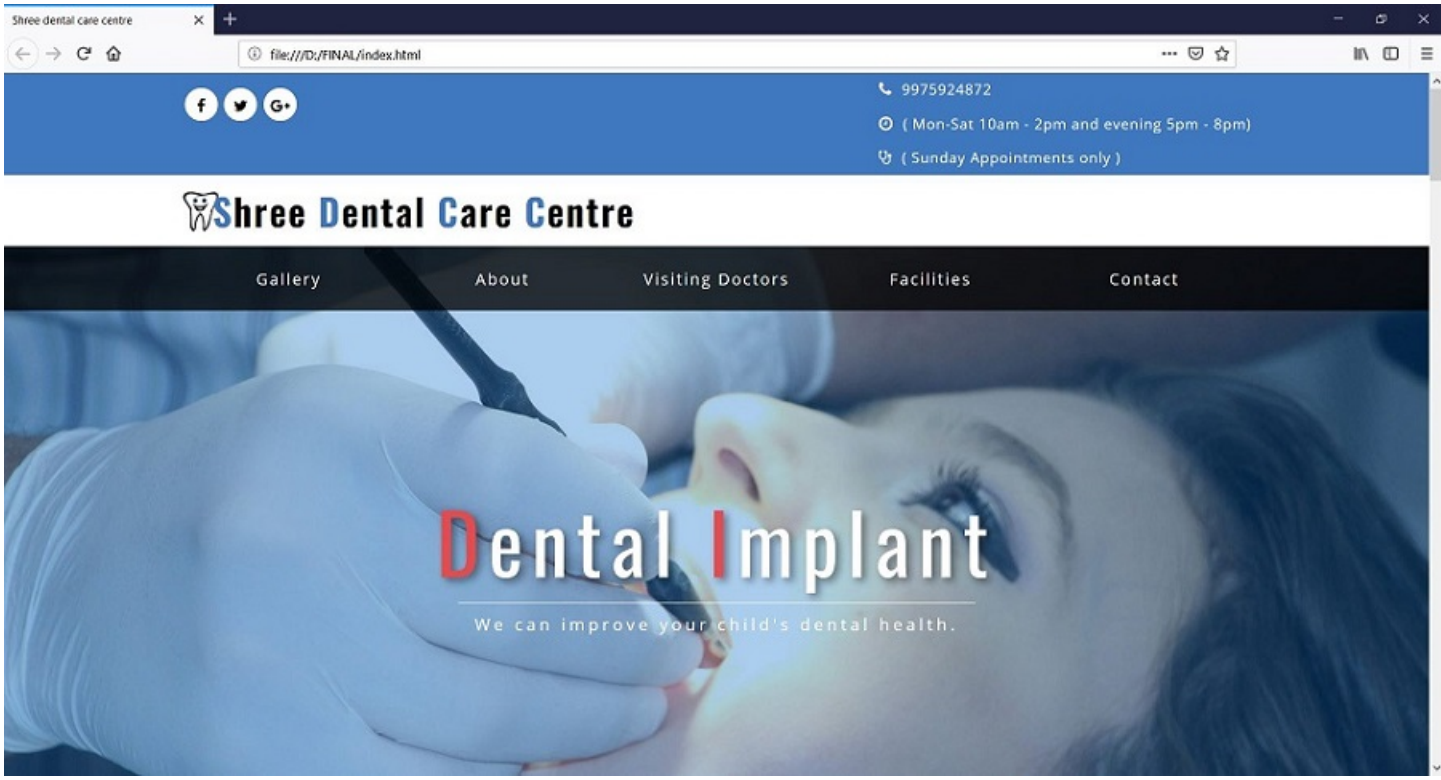

Figure 5.1: Home page of Our Web Application

ii. Appointment Form

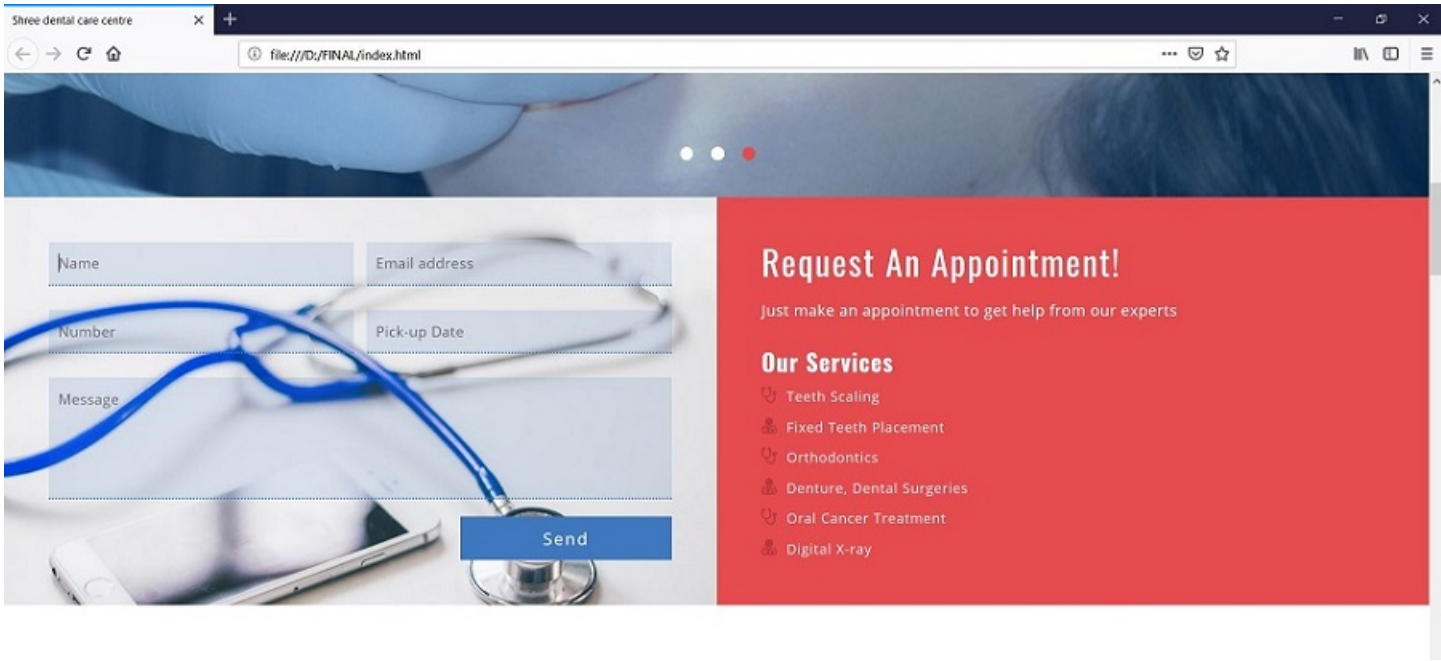

Figure 5.2: Online Appointment Application Form.

iii. About Us

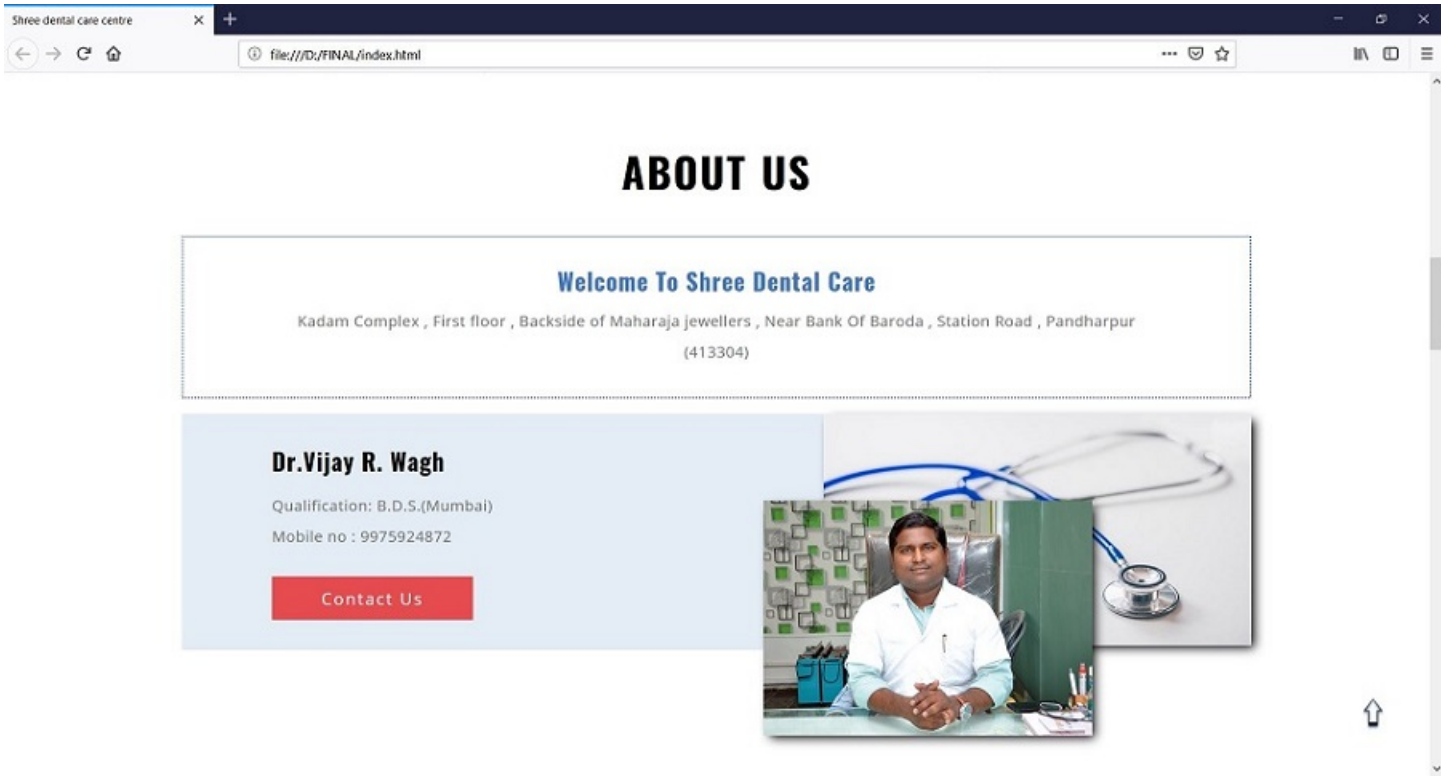

Figure 5.3: About Us Page of Web Application.

iv. Contact Us

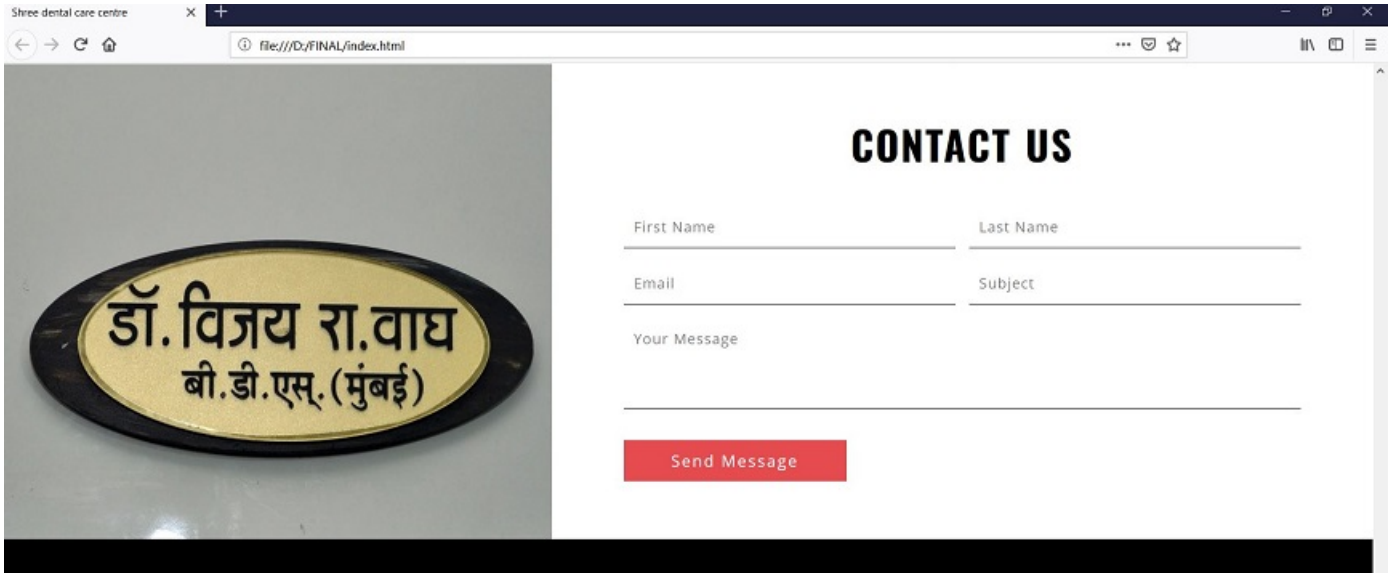

Figure 5.4: Contact Us Page of Web Application.

v. Facilities

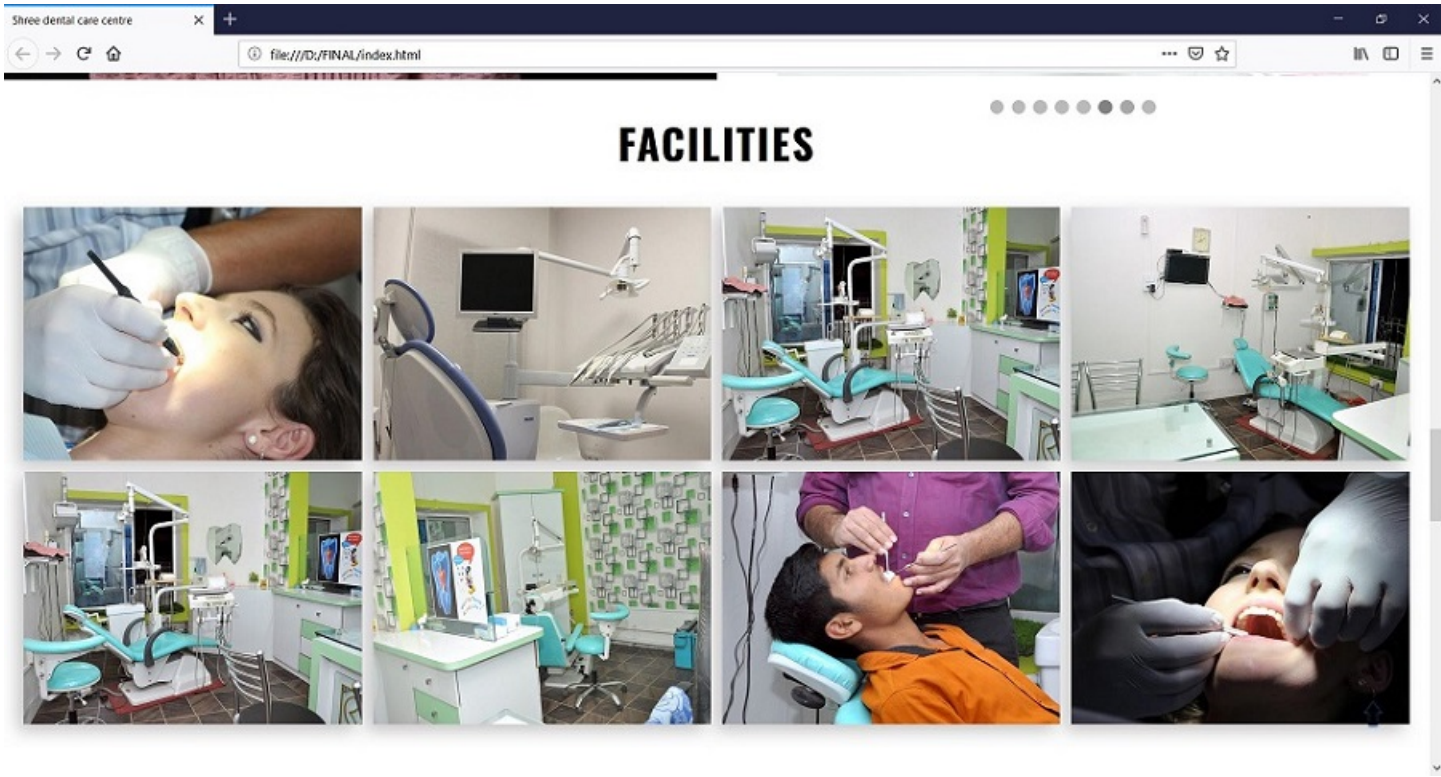

Figure 5.5: Facilities Page of Web Application.

vi. Visiting Doctors

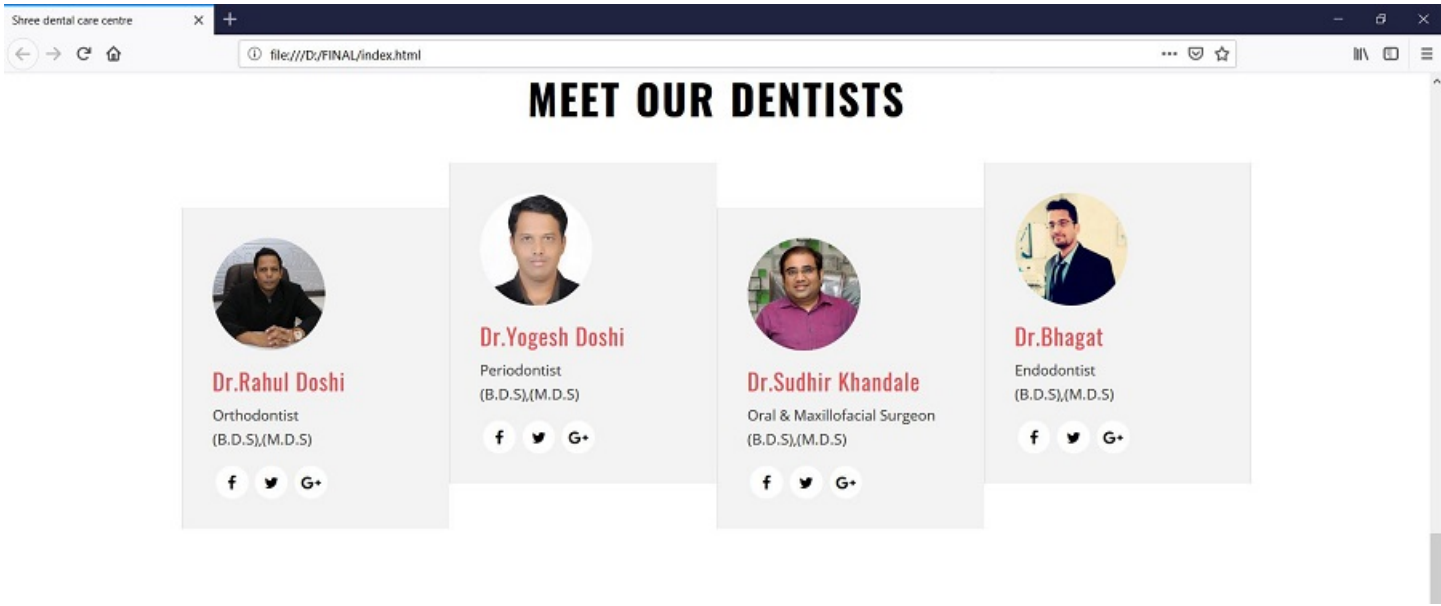

Figure 5.6: Visiting Doctors Page of Web Application.

vii. Gallery

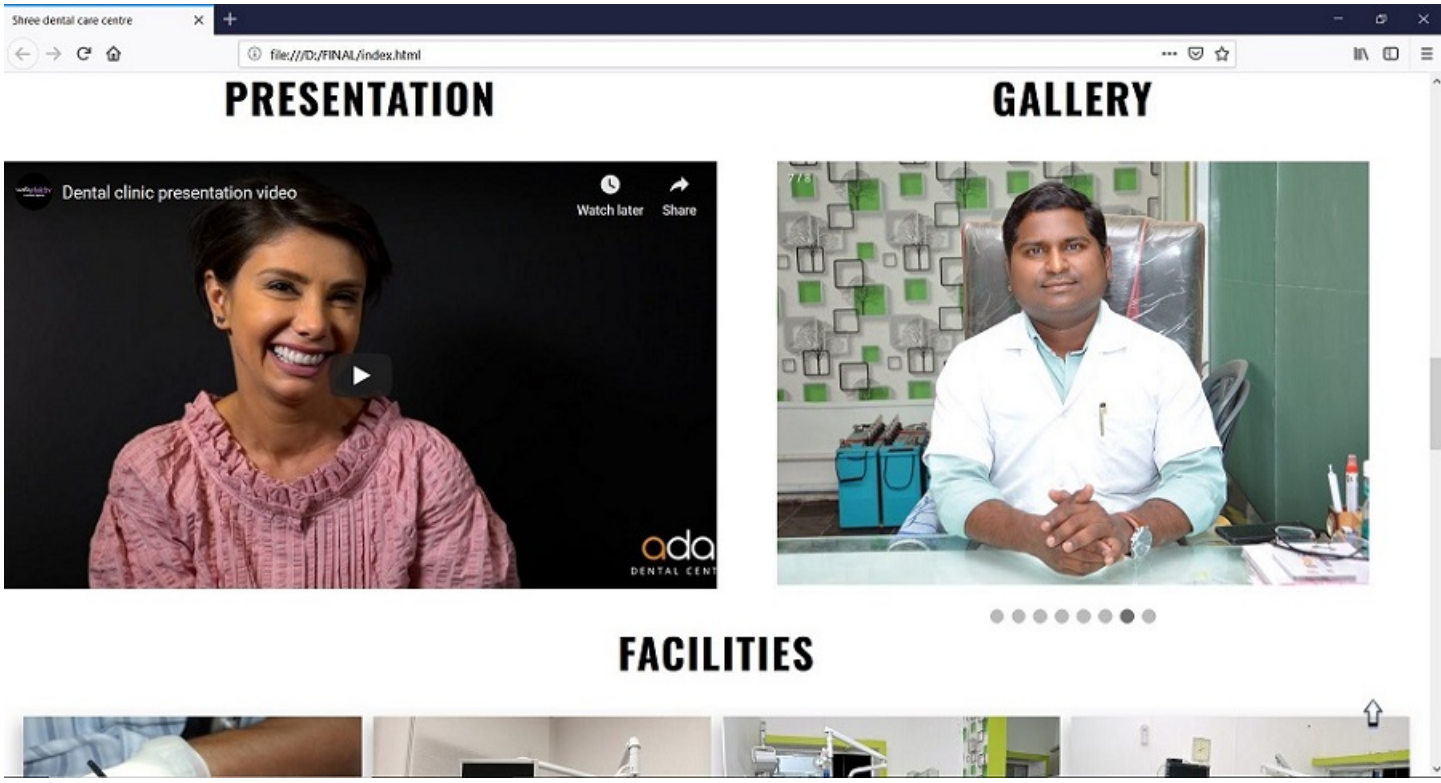

Figure 5.7: Gallery of Page of Web Application

Chapter 6

Conclusion

The project entitled as Dental Clinic System is the Web Application that deals with the issues related to a particular dental clinic. This project is successfully implemented with all the features mentioned in Application requirements specification. The application provides appropriate information to users according to the chosen service. The project is designed keeping in view the day to day problems faced by dental clinic. Deployment of our application will certainly help the dental clinic to reduce unnecessary wastage of time, paper work etc. Awareness and right information about any dental clinic is essential for both the Doctor as well as patient. So this serves the right purpose in achieving the desire requirements of both the communities.

6.1 Appendix

Acronyms	Meaning
DIMS	Dental Information Management System
DFD	Data Flow Diagram
DCS	Dental Clinic System
SDC	Shah Dental Clinic

Table 6.1: Appendix

Chapter 7

Experiment and Result Analysis

The proposed system is tested by running the system on the Mozilla Firefox and Internet Explorer with the local host server. This continuing education course is intended for general dentists, hygienists, dental assistants, dental students, dental hygiene students, and dental assistant students. This course introduces a variety of functions and concepts that facilitate the transition from a manual dental-office accounting system to a computerized system - a change that can enhance and strengthen the practice for years to come.

Bibliography

[1] Atkinson, J.C., Zeller, G.G., and Shah, C. (2002, May 1),” Electronic patient records for dental school clinics: more than paperless systems”. Journal of Dental Education. doi: 66:634-642.

[2] Greenes RA, Pappalardo AN, Marble CW, Barnett GO (1969), “Design and implementation of a clinical data management system”. Comput Biomed Res 2(5):469–485

[3] Krishnan, A., Nongkynrih, B., Yadav, K., Singh, S. and Gupta, V. (2010, November 16). “Evaluation of computerized health management information system for primary health care in rural India”. BMC Health Services Research. 2010; 10(310).

[4] Whitten J., Bentley L. and Dittman, K., [Book], System Analysis and Design Methods (5 Ed.) Mc-Graw Hill: Boston.2001, retrieved: December 19, 2011.